5TH INTERNATIONAL CONFERENCE

Smart Grid Synchronized Measurements & *Analytics*

SGSMA 2026 Smart Grid Synchronized Measurements and Analytics

Benchmarking Dynamic Frequency Estimators for Low-Inertia IBR Grids

Jorge L. Mayorga Taborda, Yessica Africano Rodriguez, Fernando Jimenez

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Date

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Venue

Santiago, Chile

Motivation

Why frequency estimation in low-inertia IBR grids is hard

Field events define the benchmark stress model

2016	2017	2021	2021	2022
Blue Cut CA, Aug. 16 **∼1,200 MW PV** inverter momentary cessation; erroneous frequency measurement on distorted fault waveforms was a contributing factor *stress: distortion → frequency artifact*	**Canyon 2** CA, Oct. 9 **∼900 MW total** 682 MW on the first fault; loss from momentary cessation, not erroneous frequency tripping *stress: PLL, voltage response, recovery*	**Odessa I/II** TX faults, 2021/22 **1,112 / 1,711 MW** remote solar reductions after normally cleared faults *stress: phase jumps, controls, model mismatch*	**Kaua'i** Nov. 21 **18–20 Hz oscillation** near or beyond ordinary 30 fps phasor-rate observability *stress: oscillation and aliasing*	**CA BESS** Mar. 9 / Apr. 6 **BESS/IBR reductions** ac overcurrent/overvoltage, dc-bus, unbalance, data-quality issues *stress: low-SNR and non-sinusoidal*

Benchmark implication: the estimator must be tested against waveform distortion, phase jumps, control recovery, unbalance, noise, harmonics, interharmonics, and reporting-rate limits, not only steady-state frequency error.

Source: NERC/WECC Blue Cut (2017); 900 MW Canyon 2 (2018); NERC Odessa I/II (2021, 2022); NERC/WECC CA BESS (2023); Dong et al., Kaua'i 18-20 Hz oscillations (arXiv:2301.05781)

Research problem

- Low-inertia IBR grids require reliable local frequency estimates during ramps, phase jumps, harmonic distortion, ringdown, and mixed dynamic events.
- Classical estimators often trade off responsiveness, noise rejection, latency, and computational load.
- Modern model-based and data-driven estimators improve some regimes, but can be difficult to compare fairly.
- Existing comparisons often under-specify seeds, disturbance families, tuning policy, trip-risk, structural latency, or computational cost.

Research gap

The literature needs a reproducible benchmark that connects estimator error to protection-oriented risk, settling behavior, and deployability.

Technical contributions

1. OpenFreqBench
reproducible stress-test platform

2. Metric-complete protocol
RMSE, peak/RoCoF error, T_{trip}, settling, CPU

3. IBR failure evidence
standard cases miss composite IBR failures

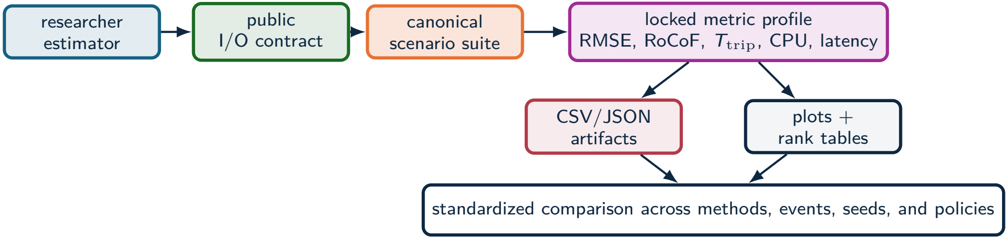
4. Extensible estimator contract
new methods enter the same scoring workflow

Contribution logic: explain estimator ranking shifts by objective, disturbance regime, and compute budget.

Benchmark & Platform

An open, reproducible estimator benchmark

OpenFreqBench platform



Researchers add

Estimator code, parameters, and hypotheses through the public contract.

Platform fixes

Truth signals, sampling, seeds, tuning policy, metric formulas, and schema.

Outputs

Benchmark reports, metric tables, plots, manifests, and readiness gates.

Principle: a submitted estimator becomes a repeatable benchmark entry under the same scenarios, seeds, metrics, and artifacts.

Experiment scale

18
estimators
loop · window · model

33
scenarios
standard + IBR

10
families
ramp → multi-event

30
seeds / pair
Monte-Carlo

18 × **33** = **594** estimator–scenario cells ▪ **594** × **30** = **17,820** Monte-Carlo records ▪ **1 MHz** truth → **10 kHz** output

Central finding: no estimator dominates — winners change with event family, metric target, and compute budget.

Source: `OpenFreqBench expanded scenario matrix`

Estimator families

Loop-based

PLL, SOGI-PLL, SOGI-FLL, Type-3 SOGI-PLL.

Model-based / stochastic

EKF, UKF, RA-EKF, LKF, LKF2.

Legacy baselines

TKEO and ZCD are included as classical, lightweight baselines.

Window / spectral

IpDFT, TFT, ESPRIT, Prony, MUSIC.

Adaptive and data-driven

RLS, Koopman (RK-DPMU).

Interpretation rule

These families occupy different latency and stress-sensitivity positions. The comparison tracks how each family behaves across the stress space.

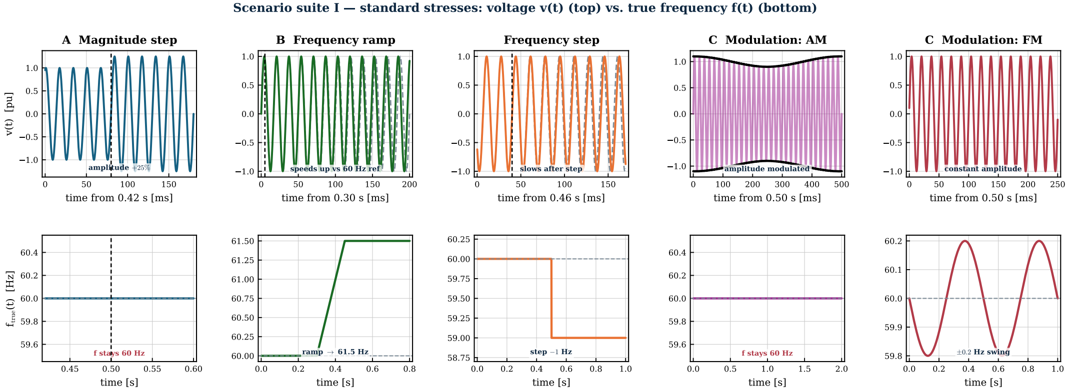
LKF and LKF2 follow H. Ahmed, S. Biricik, M. Benbouzid, "Linear Kalman Filter-Based Grid Synchronization Technique: An Alternative Implementation," IEEE, 2021.

Scenarios

Standard and composite IBR stress cases

Standard stress scenarios

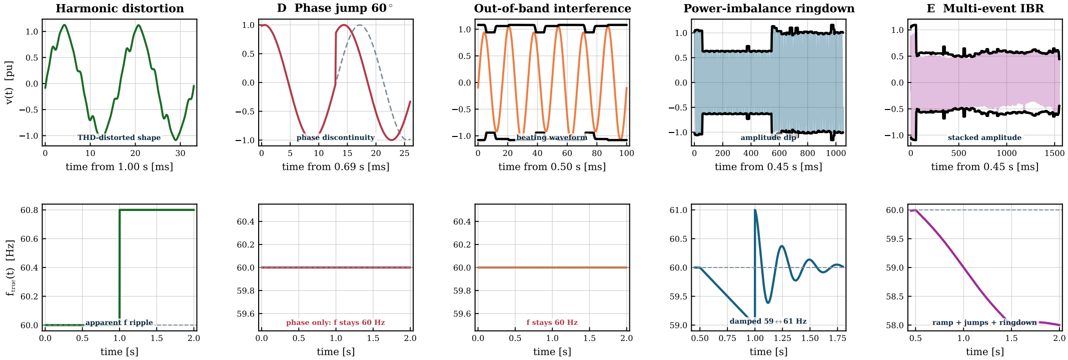
CORE SCENARIO SUITE



Composite IBR stress scenarios

IBR STRESS SUITE

Scenario suite II — composite IBR stresses: voltage v(t) (top) vs. true frequency f(t) (bottom)



Metrics & Methodology

How every estimator is measured

Experimental method

truth synthesis
1 MHz waveform
known $f(t)$, RoCoF

relay stream
10 kHz samples
local voltage only

execution
estimator API
declared tuning grid

scoring
RMSE, peak, T_{trip}
settling, CPU

evidence
tables, traces
Pareto fronts

controlled
scenario script, truth, sampling rate, seeds

input contract
single-phase voltage; no PMU time-sync assumption

fair search
each estimator–scenario pair tuned inside its declared grid

multi-objective
accuracy, recovery, protection exposure, and compute kept separate

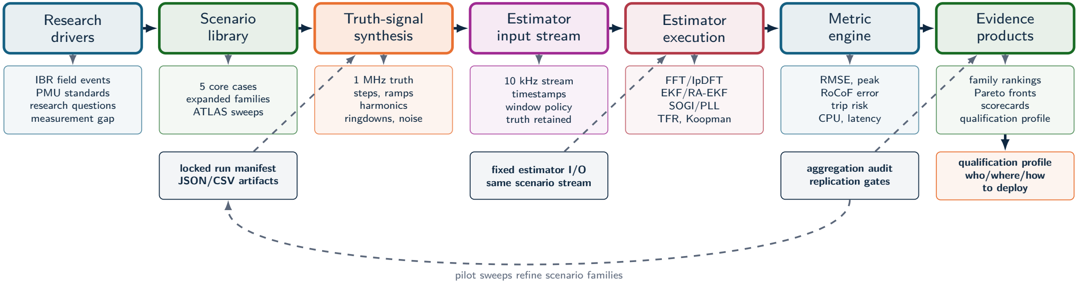
decision use
standard compliance, IBR stress, and deployment cost compared together

measurement scope
relay-class dynamic tracking

comparison rule
same truth, metrics, and timing

output rule
rank by scenario $\times$ metric $\times$ cost

Full benchmark workflow



The benchmark is an evidence pipeline, not just a test script: every figure traces back to a locked manifest, a fixed I/O contract, and the readiness gate.

Metrics and decision variables

Error and recovery

$$e[k] = \widehat{f}[k] - f_{\mathrm{true}}[k]$$

$$\mathrm{RMSE} = \sqrt{\frac{1}{N}\sum_k e[k]^2}$$

$$E_{\max} = \max_k |e[k]|$$

Protection-oriented risk

$$T_{\mathrm{trip}} = \Delta t \sum_k \mathbf{1}\{|e[k]| > 0.5\ \mathrm{Hz}\}$$

$$t_s = \min\{t : \ |e[\tau]| \leq \epsilon,\ \forall \tau \geq t\}$$

$$C_{\mathrm{CPU}} = \mathrm{mean\ time/sample}$$

The 0.5 Hz threshold is a protection-oriented risk proxy. The metric set separates average error, transient severity, trip exposure, settling, and compute cost.

Results

Winners depend on the question

Standard scenario results

Method	Step	Ramp	Mod.
RA-EKF	0.000	**0.006**	0.037
SOGI-FLL	0.005	0.013	0.040
Koopman	0.003	0.017	0.044
TFT	0.001	0.019	0.046
UKF	0.000	0.017	0.055
LKF2	0.138	0.015	**0.028**
PLL	0.033	0.073	**0.227**
IpDFT	0.000	0.037	0.114
EKF	0.000	0.010	0.064

RMSE [Hz], mean over 30 seeds. Green: below the 0.05 Hz guide; red: above it.

LKF2 implementation after H. Ahmed, S. Biricik, M. Benbouzid, "Linear Kalman Filter-Based Grid Synchronization Technique: An Alternative Implementation," IEEE, 2021.

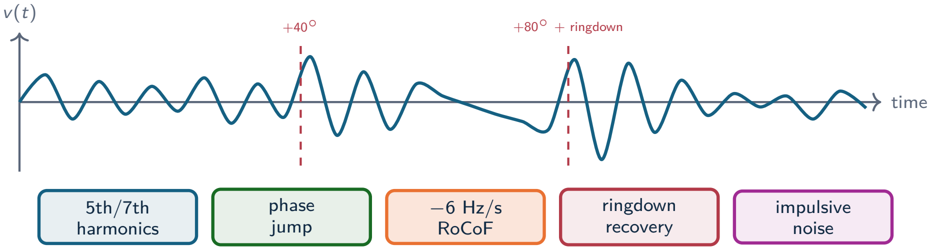
What the heatmap shows

On clean standard scenarios — magnitude step, frequency ramp, AM/FM modulation — every method tracks at or near the 0.05 Hz guide. PLL is the weakest, on modulation.

Operational consequence

Clean dynamics do not separate the methods. The real differences emerge under composite IBR stress (next slides), where leakage, re-lock, and recovery interact.

Composite IBR sequence: why it is hardest



Why standard tests miss it

The estimator sees spectral leakage, loop re-lock, state-model mismatch, transient recovery, and noise in one nonstationary waveform.

Why ATLAS decomposes it

The multi-event case shows the combined effect; severity sweeps isolate how each stressor affects each estimator family.

Scenario E: the leaderboard breaks

Method	RMSE [Hz]	T_{trip} [s]	CPU [μs]
UKF	**1.09**	**0.78**	14
EKF	1.13	0.87	11
SOGI-PLL	1.14	**0.78**	11
SOGI-FLL	1.14	0.84	**9**
IpDFT	1.14	1.35	17
RA-EKF	1.16	0.92	15
Koopman	1.90	0.84	**15 241**

No accurate winner

Under the stacked sequence every method converges to ≈ 1.1 Hz RMSE. High accuracy is simply not available in this regime.

Scalar ranking failure

UKF has the lowest RMSE and trip exposure, SOGI-FLL the lowest usable CPU, Koopman is $1000\times$ costlier; RA-EKF sits mid-pack with steady behavior.

Source: `Monte-Carlo benchmark,` $n = 30$ `seeds -- composite multi-event scenario`

Result comparison: winners depend on the question

Question	Best-supported answer	Evidence	Boundary
Clean step/modulation accuracy	State-space + SOGI-FLL near-zero; LKF2 best on mod	Step $\approx$ 0; mod 0.028–0.044 Hz	Clean cases do not predict ramp/event survival
Ramp and RoCoF tracking	RA-EKF leads; the clean ramp is tracked by all	RA-EKF 0.004 Hz; EKF/UKF/IpDFT all $<$ 0.05 Hz	RoCoF *magnitude* is the real stressor (severity curves)
Phase jump (Scenario D)	State-space filters recover with zero trip	EKF/UKF/RA-EKF $T_{\text{trip}} = 0$; IpDFT 1.21 s	Windowed/spectral carry the exposure
Multi-event RMSE	No accurate method — all $\approx$ 1.1 Hz	UKF 1.09 Hz best; wide spread across methods	RMSE hides trip-risk and 1000$\times$ CPU spread
Embedded feasibility	PLL, SOGI-FLL, EKF, RA-EKF remain plausible	9–17 μs/sample; Koopman $\sim$15,000	Python timing is a software-cost proxy

Selection changes with event family, objective metric, and compute envelope. A single leaderboard hides the deployment trade-off.

Computational feasibility

Method	Time/sample†	Cost tier
SOGI-FLL	10 μs	low
SOGI-PLL	11 μs	low
EKF	12 μs	low
RA-EKF	**15 μs**	**low**
TFT / IpDFT	12 μs	low
ESPRIT	95 μs	medium
Koopman	0.1–15 ms	high

† mean per-sample wall-clock in the Python harness (software proxy), $n = 30$; Koopman/ESPRIT scale with event length.

Deployment interpretation

CPU is a software-cost indicator from the benchmark harness, not a hardware-latency claim.

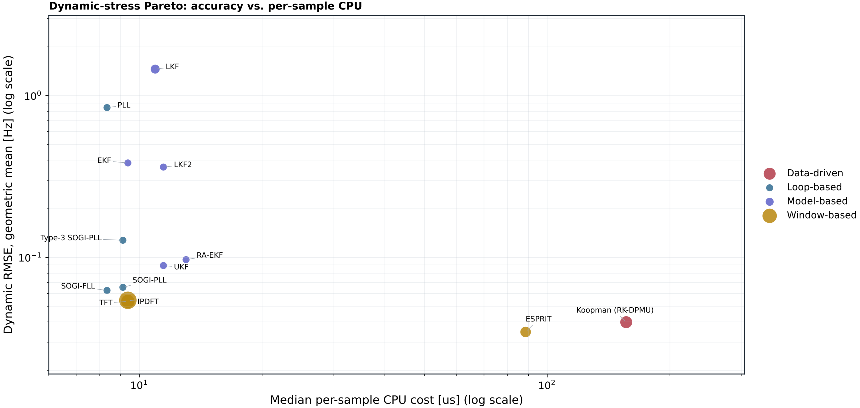
Key comparison

The practical low-cost cluster is SOGI-FLL, SOGI-PLL, EKF, and RA-EKF. Koopman and ESPRIT sit far to the right on cost without a global RMSE gain.

Accuracy costs 100–1000× more compute

OPENFREQBENCH EXPANSION: 18 estimators, 33 scenarios, $n = 30$ seeds per scenario

Deployable methods sit lower-left; spectral/data-driven accuracy moves right by 100–1000× CPU.



ATLAS severity analysis

ATLAS EXTENDED SWEEP: $n = 30$, **fixed policy, 9/9 sweeps**

Core cases → ATLAS severity curves → sign asymmetry → fixed vs oracle policy → validation gate

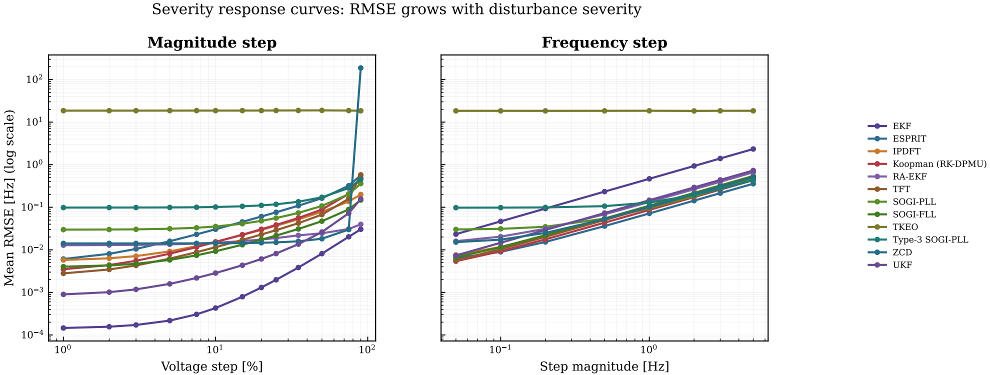
Core cases	ATLAS severity curves	sign asymmetry	fixed vs oracle policy	validation gate
A–E failure mode	When does error start to grow?	Are positive and negative events symmetric?	Does tuning policy change the rank?	Which results replicate?

ATLAS maps where estimator behavior changes with severity, sign, tuning policy, and replication depth.

Severity response: amplitude and frequency steps

ATLAS EXTENDED SWEEP: $n = 30$, fixed policy, 9/9 sweeps

Log–log RMSE vs. severity ($n = 30$). Most methods rise smoothly toward the Cramér–Rao floor; saturated curves stay high regardless of severity.

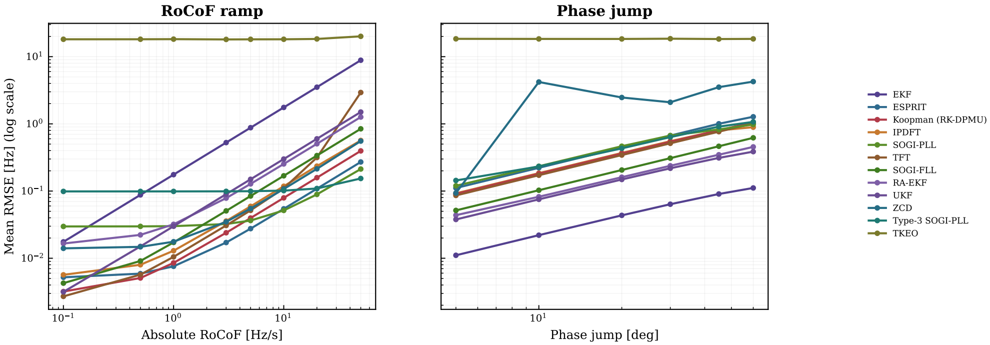


Severity response: RoCoF ramp and phase jump

ATLAS EXTENDED SWEEP: $n = 30$, fixed policy, 9/9 sweeps

Error grows with RoCoF magnitude and phase-jump angle; the curve slope and onset are the family-specific response signatures.

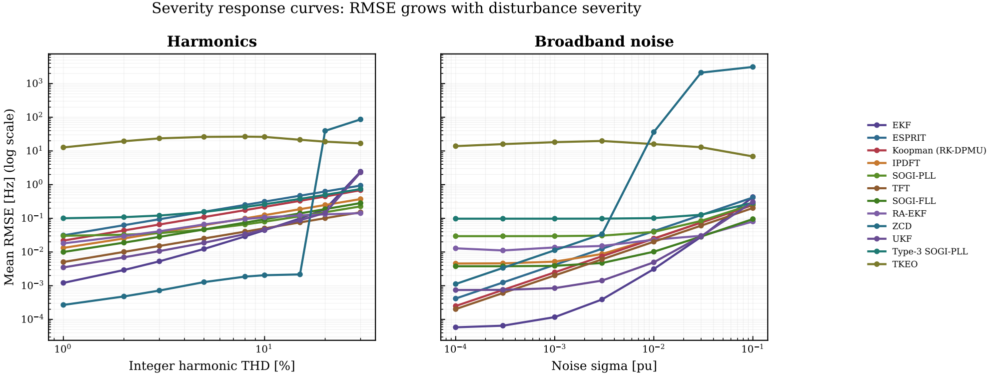
Severity response curves: RMSE grows with disturbance severity



Severity response: harmonics and broadband noise

ATLAS EXTENDED SWEEP: $n = 30$, fixed policy, 9/9 sweeps

Spectral and noise stress: several methods hold a low-error plateau, then cross a cliff into the unusable regime as severity rises.



Deployment & Conclusion

Recommendations and validation agenda

Recommendation map

FFR / RoCoF tracking

↓

EKF / RA-EKF explicit $\dot{\omega}$

Recommendation map

FFR / RoCoF tracking → EKF / RA-EKF explicit $\dot{\omega}$

Anti-islanding phase jumps → PLL / RA-EKF near-zero T_{trip}

Recommendation map

FFR / RoCoF tracking $\rightarrow$ EKF / RA-EKF explicit $\dot{\omega}$

Anti-islanding phase jumps $\rightarrow$ PLL / RA-EKF near-zero T_{trip}

Low-cost relay core $\rightarrow$ PLL / EKF CPU efficient

Recommendation map

Use case	Recommendation
FFR / RoCoF tracking	EKF / RA-EKF explicit $\dot{\omega}$
Anti-islanding phase jumps	PLL / RA-EKF near-zero T_{trip}
Low-cost relay core	PLL / EKF CPU efficient
Metering-like steady state	IpDFT / TFT if delay acceptable

Recommendation map

FFR / RoCoF tracking → EKF / RA-EKF explicit $\dot{\omega}$

Anti-islanding phase jumps → PLL / RA-EKF near-zero T_{trip}

Low-cost relay core → PLL / EKF CPU efficient

Metering-like steady state → IpDFT / TFT if delay acceptable

Deployability depends on event class, trip-risk tolerance, and compute budget.

Three findings

1. Metric-conditioned
The MC benchmark and ATLAS both show that RMSE, trip exposure, RoCoF, and CPU can select different methods.

2. Stress-bounded
AM remains survivable for many methods; severe FM, RoCoF, interharmonics, noise, and extreme THD break the 0.05 Hz guide.

3. Robustness over point accuracy
Low point error does not imply robustness: RA-EKF stays within band on every ramp seed in the tested run.

Dynamic frequency estimators should be qualified by event family, sign, severity, recovery, trip exposure, and compute envelope.

Conclusion and roadmap

Conclusion

- Field disturbances define the measurement problem; controlled synthetic truth makes estimator behavior explainable.
- The study compares estimators across objectives, regimes, and deployment constraints, including IBR-relevant stress modes.
- Empirical conclusions remain tied to evidence level, stress regime, and replication depth.

Future work — roadmap

- **Phase 1:** ANDES transient simulation of IEEE IBR standard-method fault scenarios.
- **Phase 2:** Load the simulation CSVs as benchmark truth and replay every estimator.
- **Phase 3:** HIL with FPGA estimator implementations for real-time validation.

Result: estimator choice is tied to event class, metric target, and compute envelope; estimator qualification must declare the stress regime, latency limit, and evidence depth.

Selected references

Standards, measurement & methods

- IEEE/IEC 60255-118-1:2018 — synchrophasor, frequency, and RoCoF measurement and compliance requirements.
- H. Ahmed, S. Biricik, M. Benbouzid, "Linear Kalman Filter-Based Grid Synchronization Technique: An Alternative Implementation," IEEE, 2021 *(LKF2)*.
- Open synchrophasor infrastructure: OpenPMU, openPDC, FNET/GridEye, and LBNL/PNNL open PMU event datasets.

IBR field-event reports

- NERC/WECC Blue Cut Fire (2017) and Canyon 2 Fire (2018) solar-PV interruption reports.
- NERC Odessa I/II disturbance reports (2021, 2022).
- NERC/WECC 2022 California BESS disturbance report (2023).
- Dong et al., Kaua'i 18–20 Hz oscillation analysis (arXiv:2301.05781).

Full reference list, methodology, and artifacts in the submitted paper.

Thank you

Questions & discussion welcome

Jorge L. Mayorga Taborda | Universidad de los Andes | SGSMA 2026